

Displays

D1:

$$x_{t+1} = Ax_t + Bu_t + w_t \quad (1)$$

D2:

$$X(x_0, U, W) = G_x x_0 + G_u U + G_w W, \quad (2)$$

D3:

$$r_{x_0}^U(\mathcal{S}, \mathcal{T}) = \mathbb{P}_X^{x_0, U} \{x_N \in \mathcal{T} \wedge x_t \in \mathcal{S}, \forall t \in \mathbb{N}_{[0, N-1]}\} = \mathbb{P}_X^{x_0, U} \{X \in \mathcal{R}\} 1_{\mathcal{S}}(x_0). \quad (3)$$

D4:

$$p^*(x_0) = \max_{U \in \mathcal{U}^N} r_{x_0}^U(\mathcal{S}, \mathcal{T}) \quad (4)$$

D5:

$$p^*(x_0) = \max_{U \in \mathcal{U}^N} 1_{\mathcal{S}}(x_0) \mathbb{E}_z^{x_0, U} [z], \quad (5)$$

D6:

$$\mathcal{S} = \{x | f_{\mathcal{S}} x \leq h_{\mathcal{S}}\}, \quad (6a)$$

D7:

$$\mathcal{T} = \{x | f_{\mathcal{T}} x \leq h_{\mathcal{T}}\}, \quad (6b)$$

D8:

$$\mathcal{R} = \{x | FX \leq h\} \quad (6c)$$

D9:

$$\mathcal{W}_K = \{W^{(1)}, \dots, W^{(K)}\}, \quad (7)$$

D10:

$$\begin{aligned} & \max_{U \in \mathcal{U}^N} \quad \frac{1}{K} \sum_{i=1}^K z^{(i)} \\ \text{s.t.} \quad & X^{(i)} = G_x x_0 + G_u U + G_w W^{(i)}, \quad i \in \mathbb{N}_{[1, K]}, \\ & FX^{(i)} \leq h + M(1 - z^{(i)})\mathbf{1}, \quad i \in \mathbb{N}_{[1, K]}, \\ & z^{(i)} \in \{0, 1\}, \quad i \in \mathbb{N}_{[1, K]} \end{aligned}$$

D11:

$$\lim_{K \rightarrow \infty} p_K^*(x_0) = p^*(x_0). \quad (8)$$

D12:

$$V_{\mathcal{P}}^{(j)}(\mathcal{C}) = \{p \in \mathcal{P} | d(p, c^{(j)}) \leq d(p, c^{(\ell)}), \forall j, \ell \in \mathbb{N}_{[1, \hat{K}]} \text{ and } j \neq \ell\}, \quad (9)$$

D13:

$$\text{WSS}(\hat{K}, \mathcal{V}_{\mathcal{P}}(\mathcal{C})) = \sum_{j=1}^{\hat{K}} \sum_{i=1}^K 1_{V^{(j)}}(p^{(i)}) \|p^{(i)} - c^{(j)}\|^2. \quad (10)$$

D14:

$$\mathcal{C}^* = \arg \min_{\mathcal{C} \in \mathcal{X}^{\hat{K}}} \text{WSS}(\hat{K}, \mathcal{V}_{\mathcal{P}}(\mathcal{C})), \quad (11)$$

D15:

$$\mathbb{P}_Y^K \{ \bar{Y} - \mu_{\bar{Y}} \geq \delta \} \leq e^{-2K\delta^2}. \quad (12)$$

D16:

$$K \geq \frac{-\ln(\beta)}{2\delta^2}. \quad (13)$$

D17:

$$p^*(x_0) = \mathbb{E}_z^{x_0, U^*} [z] = \frac{\mathbb{E}_Z^{x_0, U^*} [\mathbf{1}^\top Z]}{K}, \quad (14a)$$

D18:

$$p_K^*(x_0) = \frac{1}{K} \sum_{i=1}^K z^{(i)} = \frac{\mathbf{1}^\top Z}{K} \text{ under } U_K^*, \quad (14b)$$

D19:

$$\mathbb{E}_z^{x_0, U^*} [z] \geq \mathbb{E}_z^{x_0, U_K^*} [z] \quad (14c)$$

D20:

$$\mathbb{P}_Z^{x_0, U_K^*} \{p_K^*(x_0) - p^*(x_0) \geq \delta\} = \mathbb{P}_Z^{x_0, U_K^*} \left\{ \left(\frac{1}{K} \sum_{i=1}^K z^{(i)} - \mathbb{E}_z^{x_0, U_K^*} [z] \right) \geq \delta \right\}. \quad (15)$$

D21:

$$\begin{aligned} \left(\frac{1}{K} \sum_{i=1}^K z^{(i)} - \mathbb{E}_z^{x_0, U^*} [z] \right) &= \left(\frac{1}{K} \sum_{i=1}^K z^{(i)} - \mathbb{E}_z^{x_0, U_K^*} [z] \right) + \left(\mathbb{E}_z^{x_0, U_K^*} [z] - \mathbb{E}_z^{x_0, U^*} [z] \right) \\ &\leq \left(\frac{1}{K} \sum_{i=1}^K z^{(i)} - \mathbb{E}_z^{x_0, U_K^*} [z] \right) \end{aligned} \quad (16)$$

D22:

$$\mathbb{P}_Z^{x_0, U_K^*} \{p_K^*(x_0) - p^*(x_0) \geq \delta\} \leq \mathbb{P}_Z^{x_0, U_K^*} \left\{ \left(\frac{\mathbf{1}^\top Z}{K} - \frac{\mathbb{E}_Z^{x_0, U_K^*} [\mathbf{1}^\top Z]}{K} \right) \geq \delta \right\} \leq e^{-2K\delta^2}.$$

D23:

$$\begin{aligned} &\max_{U \in \mathcal{U}^N} \frac{1}{K} \sum_{j=1}^{\hat{K}} \alpha^{(j)} \hat{z}^{(j)} \\ \text{s.t. } &\hat{X}^{(j)} = G_x x_0 + G_u U + \psi^{(j)}, \quad j \in \mathbb{N}_{[1, \hat{K}]}, \\ &F \hat{X}^{(j)} \leq h - \varepsilon^{(j)} + M(1 - \hat{z}^{(j)}) \mathbf{1}, \quad j \in \mathbb{N}_{[1, \hat{K}]}, \\ &\hat{z}^{(j)} \in \{0, 1\}, \quad j \in \mathbb{N}_{[1, \hat{K}]} \end{aligned}$$

D24:

$$\phi(W) := G_w W.$$

D25:

$$\hat{X}(\psi^{(j)}) = G_x x_0 + G_u U + \psi^{(j)},$$

D26:

$$\varepsilon^{(j)} = \left[\epsilon_1^{(j)}, \dots, \epsilon_L^{(j)} \right]^\top, \quad (17)$$

D27:

$$\epsilon_\ell^{(j)} := \max_{\phi \in V_{\Phi_K}^{(j)}(\Psi_{\hat{K}})} \left(F_\ell \phi - F_\ell \psi^{(j)} \right), \quad \ell \in \mathbb{N}_{[1, L]}. \quad (18)$$

D28:

$$F_\ell \phi \leq F_\ell \psi^{(j)} + \epsilon_\ell^{(j)} \quad \forall \phi \in V_{\Phi_K}^{(j)}(\Psi_{\hat{K}}). \quad (19)$$

D29:

$$F_\ell (G_x x_0 + G_u U + \phi) \leq F_\ell (G_x x_0 + G_u U + \psi^{(j)}) + \epsilon_\ell^{(j)}. \quad (20)$$

D30:

$$F_\ell X(x_0, U, \phi) \leq F_\ell \hat{X}(x_0, U, \psi^{(j)}) + \epsilon_\ell^{(j)}. \quad (21)$$

D31:

$$\hat{p} = \frac{1}{K} \sum_{i \in \mathbb{N}_{[1, K]}} 1_{\mathcal{R}} \left(X(x_0, U_K^*, W^{(i)}) \right). \quad (22)$$

D32:

$$p_K^* \leq \hat{p} \leq p_K^*.$$

D33:

$$\ddot{x} - 3\omega x - 2\omega \dot{y} = m_d^{-1} F_x, \quad \ddot{y} + 2\omega \dot{x} = m_d^{-1} F_y. \quad (23)$$

D34:

$$\zeta_{t+1} = A\zeta_t + Bu_t + w_t. \quad (24)$$

D35:

$$\mathcal{T} = \left\{ \zeta \in \mathbb{R}^4 : |\zeta_1| \leq 0.1, -0.1 \leq \zeta_2 \leq 0, |\zeta_3| \leq 0.01, |\zeta_4| \leq 0.01 \right\}, \quad (25)$$

D36:

$$\mathcal{S} = \left\{ \zeta \in \mathbb{R}^4 : |\zeta_1| \leq \zeta_2, -1 \leq \zeta_2, |\zeta_3| \leq 0.05, |\zeta_4| \leq 0.05 \right\}, \quad (26)$$

Inline

- I1: $*$
- I2: $*$
- I3: $\mathbb{R}$
- I4: $\mathbb{N}$
- I5: $\mathbb{R}^n$
- I6: n
- I7: $\mathbb{N}_{[a,b]}$
- I8: a
- I9: b
- I10: $x \in \mathbb{R}^n$
- I11: $x^\top$
- I12: x
- I13: $\mathbf{1}$
- I14: $x_t \in \mathcal{X} = \mathbb{R}^{n_x}$
- I15: $u_t \in \mathcal{U} \subseteq \mathbb{R}^{n_u}$
- I16: $w_t \in \mathcal{W} \subseteq \mathbb{R}^{n_x}$
- I17: t
- I18: A, B
- I19: w_k
- I20: η_w
- I21: N
- I22: $X = [x_1^\top, \dots, x_N^\top]^\top \in \mathcal{X}^N$
- I23: $U = [u_0^\top, \dots, u_{N-1}^\top]^\top \in \mathcal{U}^N$
- I24: $W = [w_0^\top, \dots, w_{N-1}^\top]^\top \in \mathcal{W}^N$
- I25: G_x
- I26: G_u
- I27: G_w
- I28: x_k
- I29: X
- I30: $\mathbb{P}_X^{x_0, U}$
- I31: $\mathbb{P}_W$
- I32: $(\eta_w)^N$
- I33: $r_{x_0}^U(\mathcal{S}, \mathcal{T})$
- I34: $x_0 \in \mathcal{X}$
- I35: $U \in \mathcal{U}^N$
- I36: $\mathcal{S} \subseteq \mathcal{X}$
- I37: $\mathcal{T} \subseteq \mathcal{X}$
- I38: $\mathcal{R} = \mathcal{S}^{N-1} \times \mathcal{T}$
- I39: $z = 1_{\mathcal{T}}(x_N) \prod_{t=1}^{N-1} 1_{\mathcal{S}}(x_t) = 1_{\mathcal{R}}(X)$
- I40: $\mathbb{P}_z^{x_0, U}$
- I41: $\mathbb{P}_X^{x_0, U}$
- I42: $p^*(x_0)$
- I43: $x_0 \notin \mathcal{S}$
- I44: $\mathcal{S}$
- I45: $\mathcal{T}$
- I46: $\mathcal{R}$
- I47: $l_{\mathcal{S}}, l_{\mathcal{T}} \in \mathbb{N}, L = (N-1)l_{\mathcal{S}} + l_{\mathcal{T}}, f_{\mathcal{S}} \in \mathbb{R}^{l_{\mathcal{S}} \times n_x}$
- I48: $f_{\mathcal{T}} \in \mathbb{R}^{l_{\mathcal{T}} \times n_x}$
- I49: $F \in \mathbb{R}^{L \times n_x}$
- I50: $f_{\mathcal{S}}$
- I51: $f_{\mathcal{T}}$
- I52: $\mathcal{W}^N$
- I53: K
- I54: $p_K^*(x_0)$

I55: U_K^*
 I56: $M \in \mathbb{R}$
 I57: $W^{(i)}$
 I58: $\mathcal{W}^N$
 I59: $\mathcal{W}_K$
 I60: U
 I61: x_0
 I62: $z^{(i)}$
 I63: $FX^{(i)} \leq h$
 I64: $i \in \mathbb{N}_{[1,K]}$
 I65: $X^{(j)} \in \mathcal{R}, j \in \mathbb{N}_{[1,K]}$
 I66: $FX^{(j)} > h$
 I67: $z^{(j)} = 0$
 I68: $Z = [z^{(1)} \ \dots \ z^{(K)}]^\top$
 I69: $\{z^{(i)}\}_{i=1}^K$
 I70: Z
 I71: $\mathbb{P}_Z^{x_0, U} = \prod_{i=1}^K \mathbb{P}_z^{x_0, U}$
 I72: $\delta \in [0, 1]$
 I73: $\beta \in [0, 1]$
 I74: $\mathbb{P}_Z^{x_0, U_K^*} \{p_K^*(x_0) - p^*(x_0) \geq \delta\} \leq \beta$
 I75: $\mathbb{P}_Z^{x_0, U_K^*} \{p^*(x_0) \geq p_K^*(x_0) - \delta\} \geq 1 - \beta$
 I76: $x_0 \in \mathcal{S}$
 I77: $\hat{K} < K$
 I78: $\hat{p}(x_0)$
 I79: $\hat{p}(x_0) \leq p_K^*(x_0)$
 I80: δ
 I81: β
 I82: $\mathcal{C} = \{c^{(1)}, \dots, c^{(\hat{K})}\}$
 I83: $c^{(i)} \in \mathbb{R}^d$
 I84: $\mathcal{V}(\mathcal{C})$
 I85: $\mathbb{R}^d$
 I86: $\hat{K}$
 I87: $V^{(1)}, \dots, V^{(\hat{K})}$
 I88: $V^{(j)}$
 I89: $\forall j \in \mathbb{N}_{[1, \hat{K}]}$
 I90: $c^{(j)}$
 I91: $\mathcal{P} = \{p^{(1)}, \dots, p^{(K)}\}$
 I92: $\mathcal{V}_{\mathcal{P}}(\mathcal{C})$
 I93: $\mathcal{P}$
 I94: $\mathcal{C}$
 I95: $d(p^{(1)}, p^{(2)})$
 I96: $p^{(1)} \in \mathbb{R}^d$
 I97: $p^{(2)} \in \mathbb{R}^d$
 I98: $|V^{(j)}|$
 I99: $\mathcal{P} \in \mathcal{X}^K$
 I100: $\mathcal{C} \in \mathcal{X}^{\hat{K}}$
 I101: k
 I102: $\text{WSS}(\hat{K}, \mathcal{V}_{\mathcal{P}}(\mathcal{C}))$
 I103: $\mathcal{V}_{\mathcal{P}}(\mathcal{C}) = \{V^{(1)}, \dots, V^{(\hat{K})}\}$
 I104: $1_{V^{(j)}}(p^{(i)})$
 I105: $p^{(i)}$
 I106: $\mathcal{C}^*$
 I107: $\mathcal{V}_{\mathcal{P}}(\mathcal{C}^*)$
 I108: $\mathcal{O}(ndK\hat{K})$
 I109: $\mathcal{C}^* = \{c^{(1)}, \dots, c^{(\hat{K})}\}$

I110: $\mathcal{P} = \{p^{(1)}, \dots, p^{(K)}\}$
 I111: $\mathcal{C}_a^* = \{a + c^{(1)}, \dots, a + c^{(\hat{K})}\}$
 I112: $\mathcal{P}_a = \{a + p^{(1)}, \dots, a + p^{(K)}\}$
 I113: $p^{(i)} \in \mathcal{P}$
 I114: $p^{(i)} \in V_{\mathcal{P}}^{(j)}(\mathcal{C}^*)$
 I115: $a + p^{(i)} \in V_{\mathcal{P}_a}^{(j)}(\mathcal{C}_a^*)$
 I116: WSS
 I117: $\blacksquare$
 I118: $y^{(1)}, y^{(2)}, \dots, y^{(K)}$
 I119: $K > 0$
 I120: $y^{(i)} \in [0, 1], \forall i \in \mathbb{N}_{[1, K]}$
 I121: $\mathbb{P}_y$
 I122: $Y = [y^{(1)} \ y^{(2)} \ \dots \ y^{(K)}]^\top \in [0, 1]^K$
 I123: Y
 I124: $\mathbb{P}_Y^K = \prod_{i=1}^K \mathbb{P}_y$
 I125: $\bar{Y} = \frac{\mathbf{1}^\top Y}{K} = \frac{\sum_{i=1}^K y^{(i)}}{K}$
 I126: $\mu_{\bar{Y}} \triangleq \mathbb{E}[\bar{Y}]$
 I127: $\delta > 0$
 I128: $U_K^* \in \mathcal{U}^N$
 I129: $\mathbb{P}_Z^{x_0, U_K^*} \{p^*(x_0) - p_K^*(x_0) \geq \delta\} \leq \beta$
 I130: $U^* \in \mathcal{U}^N$
 I131: U_K^*
 I132: $\mathbb{E}_z^{x_0, U_K^*}[z]$
 I133: $\left\{ \bar{Z} \in \{0, 1\}^K : \left(\frac{\mathbf{1}^\top \bar{Z}}{K} - \mathbb{E}_z^{x_0, U_K^*}[z] \right) \geq \delta \right\}$
 I134: $\left\{ \bar{Z} \in \{0, 1\}^K : \left(\frac{\mathbf{1}^\top \bar{Z}}{K} - \mathbb{E}_z^{x_0, U_K^*}[z] \right) \geq \delta \right\}$
 I135: $\mathbb{P}\{\mathcal{S}_1\} \leq \mathbb{P}\{\mathcal{S}_2\}$
 I136: $\mathcal{S}_1 \subseteq \mathcal{S}_2$
 I137: $\mathbb{E}_z^{x_0, U_K^*}[z] = \frac{\mathbb{E}_Z^{x_0, U_K^*}[\mathbf{1}^\top Z]}{K}$
 I138: $\mathbb{P}_Z^{x_0, U_K^*} \{p_K^*(x_0) - p^*(x_0) \geq \delta\} \leq \beta$
 I139: $e^{-2K\delta^2} \leq \beta$
 I140: $K \geq \frac{-\ln(\beta)}{2\delta^2}$
 I141: p^*
 I142: p_K^*
 I143: $p_K^*(x_0)$
 I144: $\psi^{(j)}$
 I145: $j \in \mathbb{N}_{[1, \hat{K}]}$
 I146: $\phi : \mathcal{W}^N \rightarrow \mathcal{X}^N$
 I147: $\alpha^{(j)}$
 I148: j^{th}
 I149: $\sum_{j=1}^{\hat{K}} \alpha^{(j)} = K$
 I150: $\alpha^{(j)} \in \mathbb{N}_{[1, K]}$
 I151: $\varepsilon^{(j)}$
 I152: $\mathcal{W}_K$
 I153: $x_0 \in \mathcal{S}$
 I154: $\mathcal{X}_K^{x_0, U} := X(x_0, U, \mathcal{W}_K)$
 I155: $\mathcal{X}_K^{x_0, U}$
 I156: $\Psi_{\hat{K}} := \{\psi^{(1)}, \dots, \psi^{(\hat{K})}\}$
 I157: $\Phi_K := \phi(\mathcal{W}_K)$
 I158: $\phi(W) := G_w W$
 I159: $\hat{\mathcal{X}}_{\hat{K}}^{x_0, U} = \hat{X}(\Psi_{\hat{K}}) = \{\hat{X}(\psi^{(1)}), \dots, \hat{X}(\psi^{(\hat{K})})\}$
 I160: G_x

- I161: G_u
- I162: $G_x x_0 + G_u U$
- I163: $\phi(W)$
- I164: Φ_K
- I165: $\Psi_{\hat{K}}$
- I166: $\mathcal{V}_{\Phi_K}(\Psi_{\hat{K}})$
- I167: $V_{\Phi_K}^{(1)}(\Psi_{\hat{K}}), \dots, V_{\Phi_K}^{(\hat{K})}(\Psi_{\hat{K}})$
- I168: $\phi \in V_{\Phi_K}^{(j)}(\Psi_{\hat{K}})$
- I169: $FX \leq \hat{h}$
- I170: $F\hat{X}(\psi^{(j)}) \leq h - \varepsilon^{(j)}$
- I171: $\ell \in \mathbb{N}_{[1, L]}$
- I172: $F_\ell(G_x x_0 + G_u U)$
- I173: $F_\ell X(x_0, U, \phi)$
- I174: $F_\ell \hat{X}(x_0, U, \psi^{(j)})$
- I175: $F_\ell X(\phi)$
- I176: $F_\ell \hat{X}(\psi^{(j)})$
- I177: $F_\ell \hat{X}(\psi^{(j)}) \leq h_\ell - \epsilon_\ell^{(j)}$
- I178: $F_\ell X(\phi) \leq h_\ell$
- I179: $\forall \phi \in V_{\Phi_K}^{(j)}(\Psi_{\hat{K}})$
- I180: $X(\psi^{(j)})$
- I181: $F_\ell X^{(j)} \leq h_\ell - \epsilon_\ell^{(j)}$
- I182: $F_\ell X \leq h_\ell$
- I183: $\epsilon_\ell^{(j)}$
- I184: $\ell = 1, \dots, L$
- I185: $j \in \mathbb{N}_{[1, \hat{K}]}$
- I186: $\mathcal{O}(\sum_{j=1}^{\hat{K}} \alpha^{(j)} \log \alpha^{(j)})$
- I187: $\alpha^{(j)} = |V_{\Phi_K}^{(j)}(\Psi_{\hat{K}})|$
- I188: $V_{\Phi_K}^{(j)}(\Psi_{\hat{K}})$
- I189: $\hat{X}(\psi^{(j)})$
- I190: $p_{\hat{K}}^*$
- I191: p_K^*
- I192: $\phi(x_0, W) := G_x x_0 + G_w W$
- I193: $U_{\hat{K}}^*$
- I194: $\mathcal{O}(K)$
- I195: $p_{\hat{K}}^*$
- I196: U_K^*
- I197: $\hat{p}$
- I198: $U_{\hat{K}}^*$
- I199: $\hat{p} \leq p_K^*$
- I200: $U_{\hat{K}}^* = U_K^*$
- I201: $\mathcal{J} = \{j \in \mathbb{N}_{[1, \hat{K}]} | \hat{z}^{(j)} = 1\}$
- I202: $\sum_{\{i \in \mathbb{N}_{[1, K]} : W^{(i)} \in V^{(j)}\}} 1_{\mathcal{R}} \left(X(x_0, U_{\hat{K}}^*, W^{(i)}) \right) = \sum_{\{i \in \mathbb{N}_{[1, K]} : W^{(i)} \in V^{(j)}\}} 1$
- I203: $j \in \mathcal{J}$
- I204: j^{th}
- I205: $\frac{1}{\hat{K}} \sum_{j \in \mathcal{J}} \alpha^{(j)} = \hat{p}_{\hat{K}}^*$
- I206: $(z^{(j)} = 0)$
- I207: $\left(1_{\mathcal{R}} \left(X(x_0, U_{\hat{K}}^*, W^{(i)}) \right) = 1 \right)$
- I208: $\hat{p} \geq \hat{p}_{\hat{K}}^*$
- I209: $(\eta_w)^N$
- I210: $\Phi_K = \phi(\mathcal{W}_K)$
- I211: $V_{\Phi_K}^{(1)}(\Psi_{\hat{K}}), \dots, V_{\Phi_K}^{(\hat{K})}(\Psi_{\hat{K}})$
- I212: $\alpha = \{\alpha^{(1)}, \dots, \alpha^{(\hat{K})}\}$

I213: $U_{\hat{K}}^*$
 I214: $\phi(\mathcal{W}_K)$
 I215: $\varepsilon^{(1)}, \dots, \varepsilon^{(\hat{K})}$
 I216: $x, y \in \mathbb{R}$
 I217: $m_d = 300$
 I218: μ
 I219: R_0
 I220: $\omega = \sqrt{\mu/R_0^3}$
 I221: $\zeta = [x, y, \dot{x}, \dot{y}] \in \mathbb{R}^4$
 I222: $u = [F_x, F_y] \in \mathcal{U} \subseteq \mathbb{R}^2$
 I223: $w_t \in \mathbb{R}^4$
 I224: $\mathbb{E}[w_t] = 0$
 I225: $\mathbb{E}[w_t w_t^\top] = 10^{-4} \times \text{diag}(1, 1, 5 \times 10^{-4}, 5 \times 10^{-4})$
 I226: $N = 5$
 I227: $x = y = -0.75$
 I228: $\dot{x} = \dot{y} = 0$
 I229: $\mathcal{U} = [-0.1, 0.1] \times [-0.1, 0.1]$
 I230: $K = 2000$
 I231: $\mathcal{W}_N$
 I232: $\hat{K} = 20$
 I233: WSS
 I234: ζ_1
 I235: ζ_2
 I236: $p_{\hat{K}}^* \leq \hat{p} \leq p_K^*$
 I237: $\{p^*(x_0) - p_K^*(x_0) \geq \delta\}$
 I238: $\{p_K^*(x_0) - p^*(x_0) \geq \delta\}$
 I239: $\hat{p}_{\hat{K}}^*$
 I240: p
 I241: $\hat{z}^{(j)}$
 I242: $\mathcal{J}$
 I243: $\mathcal{R} = \{x | FX \leq h\}$
 I244: $F \in \mathbb{R}^{L \times n_x}$
 I245: $\forall j, \ell \in \mathbb{N}_{[1, \hat{K}]}$
 I246: $j \neq \ell$
 I247: $3\omega x$